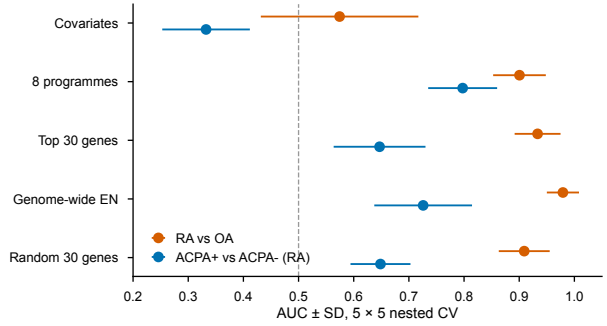
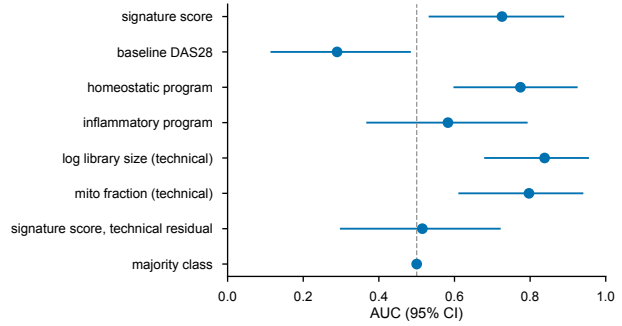
**a** **Classification against matched baselines**

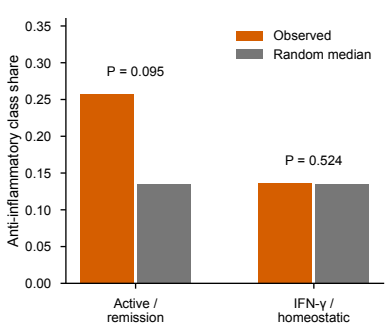



**b** **Anti-TNF response prediction and confounding**

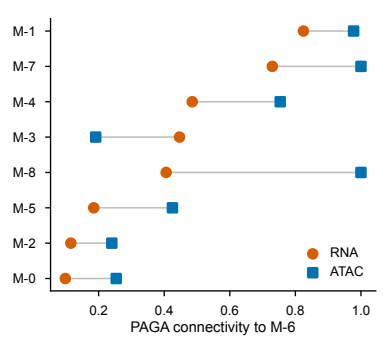



**c** **Robustness across perturbations**

**d** **Drug-class reversal versus null**




**e** **M-6 graph neighbourhood**




**f** **Boundary findings and negative controls**

**Blood monocytes**

Paired signature test: P = 0.45; 13 pairs

**Bulk deconvolution**

Failed the original sanity checks

**Active / remission classification**

AUC 0.50–0.60 in the original comparison

**RA / OA: prioritised versus random genes**

F2 vs random empirical p 0.2475

**Bulk Tph score, adjusted for T-cell content**

Random-set P = 0.81

**Myeloid S-LDSC**

Negative conditional τ z in both RA GWAS

**Trajectory timing**

Chromatin-before-RNA is not interpretable

**East Asian T-cell peak enrichment**

1.26×; P = 0.22